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High-Energy Plasma Generator Using Radio-Frequency and Neutral Beam Power

Abstract

An apparatus for generating a highly energetic plasma employs a low-energy neutral beam injected into a magnetically contained mirror plasma to produce plasma ions boosted in energy to fusion levels by a coordinated radiofrequency field.

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Background/Summary

CROSS REFERENCE TO RELATED APPLICATION [0001] This application is a continuation of U.S. patent application Ser. No. 17/995,349 filed Oct. 3, 2022 which is a US National Phase entry of PCT/US 2021/022554 which claims priority to U.S. patent application Ser. No. 16/839,780 filed Apr. 3, 2020 all of which are hereby incorporated by reference.

BACKGROUND OF THE INVENTION

[0003] The present invention relates to an apparatus for generating high-energy plasmas that can promote nuclear fusion, and in particular, to systems using magnetic mirror confinement and neutral beam injection, with additional radiofrequency power injection.

[0004] High-temperature plasmas can be confined away from a physical container and avoiding damage to the container and possible plasma quenching, by a magnetic mirror confinement system. Such confinement systems may provide an axial magnetic field extending between two ends at which the magnetic flux lines converge. Plasma ions moving within this axial magnetic field spiral along the flux lines at the local cyclotron frequency and are “reflected” by an axial component of magnetic force acting on the spiraling ions. This reflecting magnetic force caused by the flux line convergence and concomitant increasing magnetic field strength is in the direction away from the convergence. Moreover, the reflecting force is proportional to the particle kinetic energy component which is perpendicular to the magnetic field. A similar reflecting force acts on the plasma electrons.

[0005] Nuclear fusion can be promoted in a magnetic mirror confinement system by generating plasma with sufficiently high energy and density. One method of reaching this high-energy/density state injects electrically neutral particles (a neutral beam) through the magnetic containment field into the plasma where the neutral particles of the neutral beam are ionized, that is, split into plasma ions and electrons. The neutral beam has an initial energy above that necessary for fusion so that the resulting plasma ions maintain an energy suitable for fusion even with an expected collisional loss of energy of the plasma ions after introduction into the plasma. The plasma density and energy are determined by the loss rate of the fast ions injected by neutral beams which decreases with increasing beam energy; hence high energy ions are better confined than low energy ions.

[0006] A neutral beam generating a sufficient flux of highly energetic particles at energies sufficient to maintain high fusion output in a magnetic mirror confinement system is difficult and costly from an energy standpoint. Currently, such an approach does not appear to be practical for net fusion energy generation.

SUMMARY OF THE INVENTION

[0007] The present invention also injects a neutral beam injection into a magnetic mirror confinement but differs from the previous approaches by employing a low-energy neutral beam having far less energy than needed to produce significant fusion directly. Instead, after the neutral beam is ionized, the energy of those neutral beam sourced fast ions is boosted within the magnetic containment volume by using a radiofrequency electrical field. The difficulties of preferentially transferring radiofrequency energy to the fast-neutral beam ions rather than thermal ions is overcome by controlling the injection angle and energy of the neutral beam so that there is a well-defined “turning point” of the fast ions in the magnetic containment field. Tuning the radiofrequency waves to a multiple (i.e., a harmonic) of the cyclotron frequency at the turning point, preferentially energizes these neutral beam injected ions to fusion levels with only small expected wave damping effects on thermal ions.

[0008] Specifically, then, in one embodiment, the invention provides an apparatus for producing high-energy plasma in a magnetic mirror containment field, the latter providing axially-extending magnetic flux lines converging at opposed first and second ends of a containment volume holding the plasma. A neutral beam generator directs a neutral beam of particles into the containment volume at a predetermined pitch angle with respect to the magnetic field and an energy range so that the particles disassociate into plasma ions at the same pitch angle within the containment volume and have a well-defined turning point. At the turning point, fast ions have purely perpendicular energy. A radiofrequency generator can then be used to produce an electrical field to accelerate the beam-sourced ions to an energy sufficient for fusion of the plasma ions.

[0009] It is thus a feature of at least one embodiment of the invention to provide a system for boosting the energy of the plasma ions after injection into the containment field, greatly increasing the efficiency of the neutral beam.

[0010] The frequency of the electrical field may be functionally dependent on a cyclotron frequency at turning points for the plasma ions of the neutral beam in the magnetic mirror containment field.

[0011] It is thus a feature of at least one embodiment of the invention to preferentially deposit energy in the plasma ions having a matching cyclotron frequency.

[0012] In one embodiment, the frequency of the electrical field may be a harmonic of the cyclotron frequency at the turning point, greater than the cyclotron frequency.

[0013] It is thus a feature of at least one embodiment of the invention to exploit preferential transfer of radiofrequency electrical energy to resonant fast ions that occurs at higher cyclotron harmonics.

[0014] The energy of the neutral beam is set so that more than 50 percent of the neutral beam particles are converted to plasma ions.

[0015] It is thus a feature of at least one embodiment of the invention to permit the use of a lower energy neutral beam amenable to higher particle flux and thus capable of high plasma densities.

[0016] The neutral beam may have an energy of less than 50,000 electron volts.

[0017] It is thus a feature of at least one embodiment of the invention to allow setting the trade-off in the design of the neutral beam generator for high flux rates rather than high energies thereby improving ion fueling rates.

[0018] The radiofrequency generator may boost the energy of the plasma ions from the neutral beam by more than 2 times.

[0019] It is thus a feature of at least one embodiment of the invention to provide significant energy boosting of the plasma ions after injection.

[0020] The radiofrequency generator may include an antenna positioned to be proximate to a reflection limit of the plasma ions and to generate a rotating electric vector perpendicular to the axis of the magnetic mirror containment field.

[0021] It is thus a feature of at least one embodiment of the invention to optimize the antenna for energy deposition of the plasma ions.

[0022] The angle of the neutral beam may be between 15° and 80° to the axis.

[0023] It is thus a feature of at least one embodiment of the invention to provide a good trade-off between energy of the neutral beam and a turn-around point that isolates the neutral beam from thermal ions.

[0024] The apparatus may further include a treatment volume at least partially surrounding the containment volume to receive high-energy neutrons therethrough and containing an element for transmutation into a different element.

[0025] It is thus a feature of at least one embodiment of the invention to provide a system for treatment of materials with neutrons, for example, to create radiopharmaceuticals or to revitalize spent nuclear fuel.

[0026] The neutral beam may be selected from the group consisting of deuterium and tritium and in

some embodiments the system may use deuterium only with respect to the neutral beam and the gas in the containment volume.

[0027] It is thus a feature of at least one embodiment of the invention to provide a system that can work with well-understood neutral beam materials and in some cases that can avoid the use of tritium in favor of deuterium.

[0028] In one embodiment, the invention may be employed to create a fusion apparatus having a reaction volume holding a fusible material within a first axially-extending magnetic containment field. In this embodiment, a first and second plasma plug may flank the reaction volume along the axis, each plasma plug being the apparatus for producing high-energy plasma as described above, wherein plasma ions escaping from the first and second plasma plugs produce a fusion reaction in the reaction volume.

[0029] It is thus a feature of at least one embodiment of the invention to provide an improved design for a fusion device for providing transmutation or power generation.

[0030] These particular objects and advantages may apply to only some embodiments falling within the claims and thus do not define the scope of the invention.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

[0031] FIG. 1 is a perspective, cutaway view of a first embodiment of the invention providing a magnetic mirror containment field, a neutral beam generator for directing beams into the containment volume, and a radiofrequency generator producing an electric field acting on plasma ions from the neutron beam generator;

[0032] FIG. 2 is a side, elevational view of flux lines of the containment volume of FIG. 1, aligned with an end view of those flux lines, both of these views showing trajectories of plasma ions of different energies, as well as graphs of cyclotron frequency, dwell time and electric field strength as a function of axial distance; and

[0033] FIG. 3 is a simplified elevational cross-section of a fusion apparatus employing the magnetic mirror containment fields of FIG. 1 as plugs blocking the escape of high-energy plasma ions from the central solenoidal magnetic field cell.

DETAILED DESCRIPTION OF THE PREFERRED EMBODIMENT

[0034] Referring now to FIG. 1, a high-energy plasma system **10** may provide a pressure vessel **12**, for example, in the form of a sealed cylindrical shell of stainless steel or the like, extending along an axis **14** for receipt of a reaction gas, such as deuterium or tritium, through valve inlet assembly **13** from a pressure tank or the like (not shown).

[0035] First and second electromagnetic coils **16a** and **16** may be positioned within the pressure vessel **12** near the opposed ends of the pressure vessel **12** to define a containment volume **17** therebetween having a magnetic containment field **15**. The electromagnetic coils **16** are oriented and separated to form a Helmholtz pair aligned along axis **14** for establishing an axial B.sub.0 field therebetween. In one embodiment, the electromagnetic coils **16** may be pancake coils providing spirals about axis **14** powered by an external, controllable DC power supply **18** of the type understood in the art.

[0036] Positioned between the electromagnetic coils **16** but proximate to one electromagnetic coil **16b** is a radiofrequency antenna **19** (shown in simplified form), for example, providing a circularly polarized radio field extending along axis **14** when driven by a radiofrequency generator **20**. As is understood in the art, the polarized radio field provides an electrical vector **21** perpendicular to axis **14** rotating thereabout. Further discussion of loop antennas suitable for this purpose are found in T. H. Stix, "Fast Wave Heating of a Two-Component Plasma," Nuclear Fusion 15, 737 (1975) and R. W. Harvey, M. G. McCoy, G. D. Kerbel, and S. C. Chiu, "ICRF Fusion Reactivity Enhancements in

Tokamaks,” Nuclear Fusion 26, 43 (1986) hereby incorporated by reference.

[0037] A treatment volume **22** may be located radially outside the pressure vessel **12**, for example, in the form of a concentric outer cylindrical tank which may be filled with, for example, an aqueous material for transmutation by high-energy neutrons such as precursors to medical isotopes ⁹⁹Mo (molybdenum 99), ¹³¹I (iodine 131), ¹³³Xe (xenon 133), and ¹⁷⁷Lu (lutetium 177) or which may support racks holding spent nuclear fuel rods being rejuvenated through transmutation by high-energy neutrons.

[0038] A neutral beam generator **26** is positioned to inject a beam **28** of neutral particles **29** (that is non-ionized particles having zero net charge) at a pitch angle θ into the containment volume **17**. The pitch angle θ is defined as an acute angle between an angle of the beam **28** and the axis **14**. The neutral particles **29**, for example, are atoms of deuterium or tritium introduced through a gas line **24** and ionized by a local plasma (not shown). These ions are accelerated in an accelerator chamber **27** having a successive set of electrically charged plates as is generally understood in the art. The ions then pass through a neutralizing gas cell **31** to produce neutral particles **29** by a charge exchange process to produce the neutral particles **29** of the beam **28**.

[0039] Referring now also to FIG. 2, the magnetic flux lines **30** generated by the coils **16** will produce a “bottle” shape expanding radially from the axis **14** at a midpoint between the coils **16** and contracting radially at the location of the coils **16**. As is generally understood in the art, this configuration produces a mirror containment volume where randomly distributed “thermal” plasma ions of sufficient pitch angle **32** spiral around flux lines **30** between regions defined by turning points **34**.

[0040] These thermal plasma ions can be established in a variety of ways for example by using the radiofrequency antenna **19** (albeit at a low efficiency) or a separate heating system using high-frequency microwaves producing electron cyclotron resonance heating, as is understood in the art

[0041] At the regions of the turning points **34**, the thermal plasma ions **32** reverse direction caused by increasing axial components of the magnetic Lorentz force produced by the convergence of the flux lines **30**. The frequency **35** of the spiraling about the flux lines **30** is termed the “cyclotron frequency” and is a function of the strength of the magnetic field **37** along axis **14**, and for this reason the cyclotron frequency **35** generally increases toward the electromagnetic coils **16**. For ions of equal mass and charge, the cyclotron frequencies will be nominally identical at a given location along the axis **14**, independent of the velocities or energies of the ions; however, ions **32** of equal mass having different pitch angles will normally have different turning points **34**.

[0042] The velocity and hence the energy of the neutral particles **29** of the neutral beam **28** and the pitch angle θ of the neutral beam **28** are set so the majority, for example, greater than 50 percent, of the particles of the neutral beam **28** will be ionized into plasma ions **36** within the containment volume **17** before exiting the containment field. These plasma ions **36** at the same pitch angle, now having electrical charge, are captured by the magnetic flux lines **30** to increase the plasma density.

[0043] In order to promote this entrapment of the majority of the neutral particles **29** of the neutral beam **28**, the energy of the neutral beam **28** is limited to provide sufficient time-of-flight for the neutral particles **29** to be ionized. Generally, the desirable energy of the neutral beam **28** for ionization will be well below the kinetic energy required for substantial fusion, and typically less than 100 thousand electron volts or preferably less than 50,000 electron volts and more typically on the order of 15-25 keV. This is in contrast to prior art approaches which require neutral particles **29** with energies exceeding the energy necessary to promote fusion between the plasma ions **36** and typically having energies more than one million electron volts for D-D fusion. By limiting the energy of the neutral beam **28**, a trade-off may be affected in common neutron beam generators **26** to produce a higher flux density of neutral particles **29**, also increasing the plasma density.

[0044] Referring still to FIGS. 1 and 2, the pitch angle θ of the neutral beam **28** is selected to provide predetermined turning points **34'** along axis **14** for the resulting plasma ions **36** and thus to provide a corresponding predetermined cyclotron frequency **35** of the plasma ions **36** at the turning

points **34'**. This cyclotron frequency is used to set the frequency of the radiofrequency generator **20** as will be discussed below.

[0045] In addition, the antenna **19** is placed proximate to one of the turnaround points **34'** to provide a maximum field strength in that region.

[0046] Finally, within the energy levels for the neutral beam **28** that provide the desired capture of the neutral particles **29** within the containment volume **17**, the energy of the neutral beam **28** is set to be as high as possible so that the radius of orbit of the of the plasma ions **36** produced by the neutral beam **28** (gyro-orbit **52**) is higher than the average distribution gyro-orbit **52** of “thermal ions” **32**, being ions not immediately derived from the neutral beam **28**.

[0047] While the inventors do not wish to be bound by a particular theory, the above-described: (a) setting of the cyclotron frequency of the radiofrequency generator **20** to a harmonic of the cyclotron frequency of the plasma ions **36** at the turning point **34'**, (b) boosting of the energy of the plasma ions **36** above the average distribution of the thermal plasma ions **32**, and (c) maximizing the electrical field strength at the turning point **34'**, all work together to allow the radiofrequency generator **20** to preferentially boost the energy of the plasma ions **36** from the neutral beam **28** free from the damping effect of thermal plasma ions **32**.

[0048] In this regard, the setting of the radiofrequency generator **20** (per (a)) provides preferential coupling to the plasma ions **36** having a matching (e.g., a harmonically related) cyclotron frequency **35**, in contrast to thermal plasma ions **32** having a range of different Doppler-shifted cyclotron frequencies and less effective coupling. The coupling may be proportional to the square of the Bessel function $B_{n-1}(k_{\perp} v_{\perp} / \omega_{ci})$ where: [0049] n is the resonant cyclotron harmonic number of the injected wave, [0050] $k_{\perp}$ the perpendicular wave number, and [0051] ω_{ci} is the cyclotron frequency of the resonance ions.

[0052] The quantity $k_{\perp} v_{\perp} / \omega_{ci}$ may be $\sim v_{\perp} / v_A$, the Alfvén velocity of the ions (cf. T. H. Stix, “Fast Wave Heating of a Two-Component Plasma,” Nuclear Fusion 15, 737 (1975)). Given the dependence of the Bessel function on $v_{\perp}$, the coupling is proportional to powers of the perpendicular velocity of the ions, and can be adjusted to preferentially damp on hot tail ions from the neutral beam and on those diffused to higher energy by the radiofrequency waves.

[0053] Further, by setting the frequency of the radiofrequency generator **20** according to the cyclotron frequency **35** at the turning point **34'**, the influence of the electrical field from the radiofrequency generator **20** on the plasma ions **36** is increased because of the prolonged dwell time **50** of the plasma ions **36** at the turning point **34'** during their lowest axial velocity as they turn around. This is in contrast, for example, to thermal plasma ions **32** which move quickly through this zone to further turning points **34** or which do not reach as far as the turning point **34'**.

[0054] As noted above, by boosting the energy of the plasma ions **36** above the distribution of thermal plasma ions **32** (per (b)) and by setting the RF generator **20** to an RF frequency which is a high harmonic of the cyclotron frequency **35** of the plasma ions **36**, higher energy plasma ions **36** having a higher radius of gyro-orbit **52** preferentially absorb power over the thermal plasma ions **32** having a lower gyro-orbit **52**. In some embodiments, the RF frequency may be set to a range from 20 to 100 megahertz and/or to a harmonic n greater than $n=2$ and preferably $n=4$.

[0055] Generally, the higher harmonics boost the relationship between energy absorption and gyro-orbit **52** according to increasing Bessel function numbers associated with those harmonics.

Specifically, energy absorption will be proportional to $J_{n-1}(k_{\perp} \rho)$ where: J_{n-1} is the Bessel coefficient for a given harmonic n , ρ is the radius of the particle's gyro-orbit **52** about the magnetic flux lines **30** which increases with energy by

$$[00001] \quad \rho = \frac{\sqrt{2} m E}{e B}$$

and $k_{\perp}$ is a wave number of the plasma ions **36** being a property of the wave within the plasma and the polarization of the antenna **19** launching the wave.

[0056] It will be appreciated that this effective preferential absorption of energy by the plasma ions

36 will be self-reinforcing as energy is absorbed and the gyro-orbit of the plasma ions **36** is increased.

[0057] Finally, by placing a highest field strength of the antenna **19** near the turning point **34'**, the plasma ions **36** are preferentially influenced.

[0058] Generally, the magnetic containment field **15** will tend to lose some plasma ions **32** having low pitch angles through its ends. These particles are said to be in the “loss cone.” By boosting the population of the plasma ions **36** having a known pitch angle θ outside of the loss cone, increased plasma densities can be obtained.

[0059] While the cyclotron frequency of the plasma ions **36** near the turning point **34'**, and hence the desired setting of the frequency of the radiofrequency generator **20** is primarily a function of the vacuum magnetic field strength **37**, it will shift slightly as a function of increasing plasma density/pressure. Accordingly, the invention contemplates that either or both of the DC power supply **18** or the RF frequency generator **20** may be adjusted during operation to maintain the above relationships which boost energy transfer to the plasma ions **36**. In particular, this adjustment may be made via a closed-loop feedback control using a sensor **56** detecting plasma pressure, for example, using a diamagnetic loop, which will measure the decrease in magnetic field due to increased plasma pressure to ensure a matching of the excitation frequency of the RF generator **20** with the actual and dynamic cyclotron frequency **35** at the turnaround point **34'**. To the extent that the cyclotron frequency is dictated by the total field (vacuum field from coil plus plasma diamagnetism); the invention also contemplates that no frequency change may be required but the location of the turning point will move closer to the electromagnetic mirror coil.

[0060] Referring now to FIG. **3**, this benefit of the present invention in providing high plasma densities makes it useful as part of a system where two high-energy plasma systems **10** may act as “plugs” to trap high-energy plasma ions in a larger scale neutron generator **60** for the purpose of transmutation (as discussed above) or fusion power generation. Such a design, for example, may make use of a tandem mirror scheme, for example, described at G. Dimov, V. Zakaidakov, and M. Kishinevski, *Fiz. Plazmy* 2 597 (1976), [*Sov. J. Plasma*], *Phys* 2, 326 (1976)] and T. K. Fowler and B. G. Logan, *Comments on Plasma Physics and Controlled Fusion* 2, 167 (1977) and hereby incorporated by reference.

[0061] More specifically, in such a tandem mirror neutron generator **60**, first and second high-energy plasma systems **10a** and **10b** are placed in opposition along axis **14** flanking a generator volume **62**. Generally, the high-energy plasma systems **10** will have an axial length on the order of 2 meter whereas the generating volume **62** will be much larger, for example, on the order of 50 meters or more.

[0062] The electromagnetic coils **16** of both of the high-energy plasma systems **10a** and **10b** are axially aligned to provide a same direction of polarization of the magnetic field along the common axis **14**. As such, the flux lines **30** of the first high-energy plasma system **10a** may continue through the volume **62** to the second high-energy plasma system **10b**. Within the volume **62**, the flux lines **30** are focused by an axially-extending solenoid coil **66** circling the axis **14** around the volume **62**.

[0063] For this purpose, the electromagnetic coils **16** may be superconducting magnets for example per D. Whyte, J. Minervini, B. LaBombard, E. Marmar, L. Bromberg, and M. Greenwald, “Smaller and sooner: Exploiting high magnetic fields from new superconductors for a more attractive fusion energy development path,” *Journal of Fusion Energy*, 35, 41 (2016) also hereby incorporated by reference.

[0064] A subset of thermal plasma ions **32**, having a uniform distribution of pitch angles and having been boosted to higher energies by kinetic transfer from the plasma ions **36**, may escape from the high-energy plasma systems **10** into the volume **62** containing a reactant gas, for example, deuterium or tritium, to promote fusion and the emission of neutrons **64** from the volume **62**. The high pressure of the high-energy plasma systems **10** blocks the escape of high-energy plasma ions from the volume **62** to maintain the high densities for significant fusion.

[0065] The volume 62 may be surrounded by a contained volume 22 which may include a heat exchanger liquid 68, for example, for receiving, through one or more heat exchangers, a working fluid 70 of a thermodynamic engine such as a turbine or the like, for example, for the generation of electrical power. Alternatively, the contained volume 22 may be used for the transmutation of materials to generate medical isotopes or to rejuvenate spent nuclear fuel as discussed above.

[0066] The present application incorporates disclosure of US patent application 2019/0326029 entitled: Apparatus and Method for Generating Medical Isotopes, and US application 2013/0142296 entitled: Apparatus and method for generating medical isotopes which describe additional techniques for managing isotope transmutation including the use of neutron multiplier generators and other construction details and mechanisms for producing a neutral beam discussed above.

[0067] Certain terminology is used herein for purposes of reference only, and thus is not intended to be limiting. For example, terms such as “upper”, “lower”, “above”, and “below” refer to directions in the drawings to which reference is made. Terms such as “front”, “back”, “rear”, “bottom” and “side”, describe the orientation of portions of the component within a consistent but arbitrary frame of reference which is made clear by reference to the text and the associated drawings describing the component under discussion. Such terminology may include the words specifically mentioned above, derivatives thereof, and words of similar import. Similarly, the terms “first”, “second” and other such numerical terms referring to structures do not imply a sequence or order unless clearly indicated by the context.

[0068] When introducing elements or features of the present disclosure and the exemplary embodiments, the articles “a”, “an”, “the” and “said” are intended to mean that there are one or more of such elements or features. The terms “comprising”, “including” and “having” are intended to be inclusive and mean that there may be additional elements or features other than those specifically noted. It is further to be understood that the method steps, processes, and operations described herein are not to be construed as necessarily requiring their performance in the particular order discussed or illustrated, unless specifically identified as an order of performance. It is also to be understood that additional or alternative steps may be employed.

[0069] It is specifically intended that the present invention not be limited to the embodiments and illustrations contained herein and the claims should be understood to include modified forms of those embodiments including portions of the embodiments and combinations of elements of different embodiments as come within the scope of the following claims. All of the publications described herein, including patents and non-patent publications, are hereby incorporated herein by reference in their entireties.

Claims

1-17. (canceled)

18. An apparatus for producing high-energy plasma, comprising: magnets configured to provide a magnetic mirror containment field with axially-extending magnetic flux lines converging proximate to a first axial end and a second axial end of a volume containing a plasma; a neutral beam generator configured to direct a neutral beam of particles into the plasma to generate ions in the volume, the generated ions having a first energy less than a fusion energy threshold of the generated ions; and a radiofrequency generator configured to produce an electric field, the electric field configured to accelerate the generated ions in the containment volume to a second energy greater than or equal to the fusion energy threshold of the generated ions.

19. The apparatus of claim 18, wherein the first energy is less than about 50 keV.

20. The apparatus of claim 18, wherein an energy of the neutral beam is less than about 50 keV.

21. The apparatus of claim 18, wherein at least a portion of the generated ions travel along the axially-extending magnetic flux lines.

22. The apparatus of claim 21, wherein the convergence of the axially-extending magnetic flux lines proximate to the first axial end and the second axial end generates an axial magnetic force on the generated ions, the axial magnetic force directed towards a center of the volume.
23. The apparatus of claim 22, wherein the axial magnetic force on the generated ions creates turning points proximate to the first axial end and the second axial end such that the generated ions reverse a direction of travel proximate to the first axial end and the second axial end.
24. The apparatus of claim 18, wherein the radiofrequency generator includes: a radiofrequency source configured to direct a radiofrequency wave into the plasma at a predetermined frequency to selectively heat at least a portion of the generated ions.
25. The apparatus of claim 24, wherein the radiofrequency source is a first radiofrequency source, the radiofrequency wave is a first radiofrequency wave, and the predetermined frequency is a first predetermined frequency, the apparatus further comprising: a second radiofrequency source configured to direct a second radiofrequency wave into the plasma at a second predetermined frequency different from the first predetermined frequency.
26. The apparatus of claim 25, wherein the second predetermined frequency is greater than the first predetermined frequency such that the second radiofrequency source is configured to selectively heat electrons in the plasma.
27. The apparatus of claim 18, wherein the plasma includes thermal ions having an energy less than the first energy of the generated ions.
28. The apparatus of claim 18, wherein the frequency of the electrical field is dependent on a cyclotron frequency at turning points for the generated ions in the magnetic mirror containment field.
29. The apparatus of claim 28, wherein the frequency is a harmonic of the cyclotron frequency that is greater than the cyclotron frequency.
30. The apparatus of claim 18, wherein the energy of the neutral beam is configured so that more than 50 percent of the neutral beam particles are converted to the generated ions.
31. The apparatus of claim 18, wherein the radiofrequency generator boosts the energy of the generated ions by more than 2 times.
32. The apparatus of claim 18, wherein the neutral beam has a pitch between about 30° and about 60° to a longitudinal axis of the volume.
33. A fusion apparatus, comprising: a reaction volume configured to hold a fusible material within a first axially-extending magnetic containment field; a first plasma plug and a second plasma plug flanking the reaction volume along the axis, each of the first and second plasma plugs including: (a) magnets configured to provide a magnetic mirror containment field converging at opposed first and second ends of a containment volume holding a plasma; (b) a neutral beam generator configured to direct a neutral beam of particles into the plasma to generate ions in the containment volume, the neutral beam having a first energy less than a fusion energy of the generated ions; and (c) a radiofrequency generator configured to produce an electric field, the electric field configured to accelerate the generated ions in the containment volume to a second energy greater than the first energy.
34. The fusion apparatus of claim 33, wherein the reaction volume contains a reaction plasma including plasma ions, and a pressure of the generated ions within the first and second plasma plug is greater than a pressure of the plasma ions within the reaction volume.
35. The fusion apparatus of claim 33, further comprising: a contained volume including a fluid, the contained volume coupled to the reaction volume and configured to receive neutrons from the reaction volume to heat the fluid.
36. The fusion apparatus of claim 35, further comprising: a heat exchanger operably coupled to the contained volume, the heat exchanger configured to convert heat received from the heated fluid to electrical energy.
37. An apparatus for producing high-energy plasma, comprising: magnets configured to generate a

magnetic mirror containment field with axially-extending magnetic flux lines converging at opposed first and second ends of a volume holding a plasma; a first energy source configured to direct a first energy into the plasma, the first energy less than a fusion energy of ions in the plasma; and a second energy source different from the first energy source, the second energy source configured to accelerate the ions to a second energy, the second energy greater than or equal to the fusion energy of the ions.

38. The apparatus of claim 37, wherein the first energy source is a neutral beam generator configured to direct a neutral beam of particles at the first energy into the plasma.

39. The apparatus of claim 37, wherein the second energy source is a radiofrequency generator configured produce an electrical field configured to accelerate the ions to the second energy.

40. The apparatus of claim 37, wherein the first energy is less than or equal to 50 keV.

41. A method, comprising: containing a plasma in a volume with a magnetic mirror containment field; directing a neutral beam of particles into the plasma at a first energy to generate ions in the plasma, the first energy less than a fusion energy threshold of the generated ions; and accelerating the generated ions to a second energy greater than or equal to the fusion energy threshold of the generated ions via an electrical field generated by a radiofrequency wave.

42. The method of claim 41, further comprising: initiating a fusion reaction in the reaction volume via the generated ions.

43. The method of claim 41, wherein the first energy is less than or equal to 50 keV.

44. The method of claim 41, further comprising: generating the magnetic mirror containment field via magnets, the magnetic mirror containment field having axially-extending magnetic flux lines converging at opposed first and second ends of the volume.

45. The method of claim 41, wherein the radiofrequency wave has a predetermined frequency that is a harmonic of a cyclotron frequency of the generated ions.

46. The method of claim 45, wherein the radiofrequency wave is a first radiofrequency wave, and the predetermined frequency is a first predetermined frequency, the method further comprising: heating electrons in the plasma via a second radiofrequency wave having a second predetermined frequency greater than the first predetermined frequency.
